

TECHNICAL SPECIFICATION GUIDE

Product Research Assistant Development, Latency Debugging, and Production Setup

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Target Services: Google Cloud Run (mcp-server & backend)

Client SDKs: Google ADK & FastMCP SSE

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1. System Design: The 'Why' and the 'How'

Consumer electronics research is typically hindered by fragmented specifications, pricing variability, and hard-to-parse comparisons. The Product Research Assistant was built to bridge this gap. By structuring a client-agent-tool lifecycle, the application provides a zero-friction dashboard capable of returning optimized recommendations under 15 seconds.

1.1. Core Requirements & Engineering Motives

- **Zero Local Dependencies:** Serves a fully compiled single-page React app directly out of the FastAPI static mounts, ensuring client-side asset portability.
- **Fast Tool Routing:** Decouples tool execution to a dedicated MCP server. By separating database lookups from LLM processing, the agent stays lightweight and decoupled.
- **Latency Mitigation:** Migrated the model from legacy/deprecated definitions to *gemini-3.5-flash-lite*, yielding a 92% latency reduction (from 209s down to 14s).

1.2. Decoupled Service Topology Diagram

Layer / Service	Technology	Role & Communication Details
Client UI Dashboard	React 18, Tailwind CSS, Vite	Renders the chat console, comparison tables, and clickable manufacturer URL
Web & Agent Router	FastAPI, Uvicorn, ADK Agent	Hosts session mappings, serves React static files, routes prompts to Gemini m
Tool Registry	Model Context Protocol (SSE)	Exposes python functions as SSE-based tools to the ADK Agent network.
Product Catalog	SQLite/In-memory Python DB	Stores 26 products with Pydantic specs, pros, cons, and manufacturer links.

2. Detailed Implementation History

To upgrade the prototype into a universal electronics research engine, the stack was refactored as follows:

2.1. Adding Manufacturer URLs & Expanded Database Schema

A string-typed *url* parameter was added to the Product model definition inside *mcp_server/server.py*. The mock database catalog was loaded with 26 high-tier items spanning smartphones, tablets, laptops, audio systems, and gaming screens.

2.2. Category Keyword Synonyms Routing Logic

To enable matching for broad requests (like 'I need a tablet'), the search tool was refactored with the following mapping logic:

```
if not matches_query: category_mapping = { "laptop": ["laptop", "notebook", "computer",
"programming", "coder", "developer"], "tablet": ["tablet", "tab", "ipad"], "smartphone":
["phone", "smartphone", "iphone", "pixel", "galaxy", "mobile"], "audio": ["audio", "headphones",
"earbuds", "noise canceling", "anc", "music"], "monitor": ["monitor", "display", "screen"] }
product_category = product.specs.get("Category", "").lower() for cat, keywords in
category_mapping.items(): if product_category == cat and any(kw in query_lower for kw in
keywords): matches_query = True break
```

This logic automatically resolves category intents, matching items by checking if their category metadata aligns with the search keywords.

2.3. Frontend Hyperlinks Integration

The React single-page frontend layout was updated to map product links using Lucide-react's *ExternalLink* icon:

- **Result Cards:** Injected a secondary styled link button below each product's pros/cons list pointing to *product.url*.
- **Best Choice View:** Placed a primary cyan-colored CTA button 'View Official Product Page' referencing the selected candidate's URL.
- **Comparison Table:** Appended 'Product Page' directly to the comparison dictionary, generating clickable 'View Page' hyperlinks inside the specification matrix.

3. Local Run & Deployment Guide

Navigate and execute the following scripts in separate terminals to spin up local environments:

```
# Terminal 1: Launch the MCP tool registry cd mcp_server ..\backend\.venv\Scripts\python
server.py 8081 # Terminal 2: Spin up the FastAPI agent container cd backend .venv\Scripts\python
main.py
```

4. Production Verification Trace Logs

Below is the exact output trace from a live deployment check, validating category search routing and URL injection:

```
POST /run with payload 'I need a tablet under $500' Status Code: 200 OK Received 9 events. [Tool
Call] product__search_products: {'query': 'tablet', 'max_price': 500} [Tool Response] matched:
Galaxy Tab A9+ ($219.00), iPad 10th Gen ($349.00) [Tool Call] product__compare_products:
['samsung-galaxy-tab-a9-plus', 'apple-ipad-10th-gen'] Streaming response: - Recommendation:
Apple iPad 10th Gen ($349.00) - Link: https://www.apple.com/ipad-10.9/ - Alternative: Galaxy Tab
A9+ ($219.00) - Link: https://www.samsung.com/us/tablets/galaxy-tab-a9-plus/
```